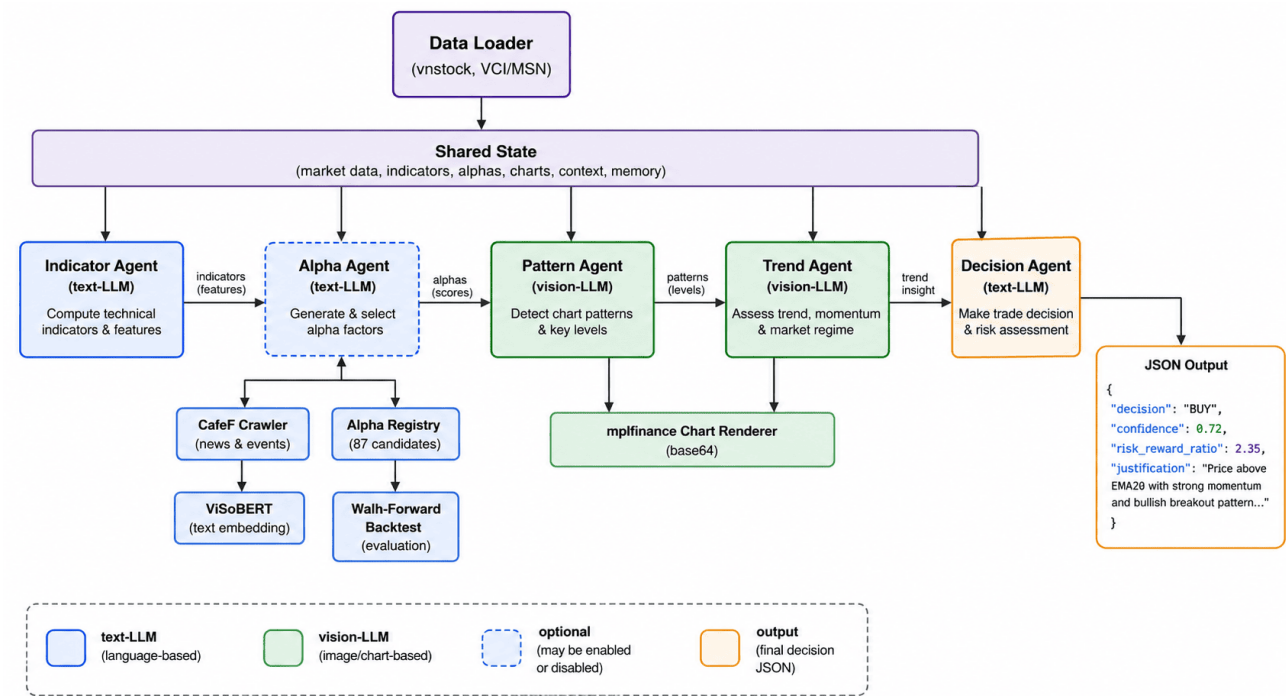


Graphical Abstract

A Multi-Agent Large Language Model Approach to Price-Driven Stock Trading in the Vietnamese Stock Market*

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Highlights

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- A leakage-safe five-agent LangGraph system for Vietnamese equities.
- Paired shared reports isolate the Alpha bundle on 180 test points.
- Full accuracy reaches 65.0% versus 51.1% for No-Alpha.
- A compounded long-only ledger separates accuracy from economic return.
- Four-way ablation disentangles alpha factors and sentiment context.

A Multi-Agent Large Language Model Approach to Price-Driven Stock Trading in the Vietnamese Stock Market

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Abstract

Algorithmic trading research increasingly uses large language models (LLMs), yet most systems assume liquid, English-dominated markets. We present a five-agent LangGraph system for Vietnamese equities that combines technical indicators, point-in-time selection from 85 OHLCV alpha candidates, Vietnamese sentiment context, vision-language chart analysis, and a binary LONG/SHORT decision. A paired shared-report protocol holds Indicator, Pattern, and Trend evidence fixed between Full and No-Alpha branches, while a fixed-cycle long-only ledger compounds wealth and charges fee and slippage on both transaction legs. Across nine equities and 180 paired test points, Full accuracy is 65.0% versus 51.1% for No-Alpha, a mean lift of 13.9 percentage points (cross-symbol Wilcoxon $p = 0.0078$). Mean Full account return remains negative at -1.69% , so directional improvement does not imply profitability. A four-way ablation on FPT, VNM, and VCB attributes a larger independent contribution to alpha factors ($+8.33$ pp on average) than sentiment ($+1.67$ pp) and reveals heterogeneous interactions.

Keywords: Multi-agent LLM, Vietnamese stock market, Alpha factor selection, Sentiment analysis

1. Introduction

Large language models (LLMs) have recently shown strong potential as reasoning engines in financial analysis, enabling multi-agent systems to coordinate over heterogeneous signals-price data, technical indicators, chart patterns, and news sentiment-within a unified pipeline. While frameworks such as QuantAgent [1] establish a compelling blueprint for LLM-driven trading agents, they are designed for mature, English-language markets and do not generalize to the constraints of emerging economies.

Vietnam’s equity market presents three challenges that require explicit adaptation. First, a predictive bar horizon and the assumptions of an executable cash-equity simulation are distinct; calendar gaps and instrument scope make them non-interchangeable. Second, the dominant financial news source, CafeF, publishes exclusively in Vietnamese, making English-centric sentiment pipelines inapplicable. Third, the shallow cross-sectional universe of HOSE, HNX, and UPCoM calls for alpha factor selection methods suited to single-stock Open, High, Low, Close, and Volume (OHLCV) data rather than the large-universe assumptions of standard factor libraries.

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We present a five-agent LangGraph [2] pipeline that addresses each of these gaps. An Alpha Agent dynamically selects the top-five factors from an 85-candidate registry-adapting the full WorldQuant 101 corpus [3] to single-stock data-ranked by walk-forward Information Coefficient (IC), accuracy, and Sharpe ratio. The factor values and ranking use OHLCV only. CafeF sentiment is supplied separately as textual context for qualitative LLM comparison. The primary configured scorer is a hosted ViSoBERT [4] endpoint, with a deterministic lexicon fallback; the scorer and fallback status are recorded per article, and a run is attributed to the endpoint only when its included articles verify that path. Pattern and Trend agents apply vision-language analysis to candlestick and trendline charts, while a Decision Agent synthesises all reports into a binary LONG/SHORT direction forecast whose account action is resolved separately.

The primary objective of this paper is to evaluate whether the Alpha Agent improves directional forecasting relative to a no-alpha baseline. Indicator, Pattern, and Trend reports are generated once at each cutoff and cloned into both decision branches, providing paired common-support estimates without upstream variation inside a comparison. The nine-symbol benchmark reports directional, statistical, and compounded-account outcomes. A separate four-way shared-upstream experiment then evaluates Full, Alpha-only, Sentiment-only, and Baseline variants, allowing the factor and sentiment contributions to be distinguished rather than treating the Alpha bundle as indivisible.

The remainder of this paper is organised as follows. Section 2 reviews related work on LLM-based trading agents, Vietnamese NLP, and quantitative alpha research. Section 3 formalises the problem. Section 4 describes the system architecture. Section 5 presents the experimental setup. Section 6 reports backtest results. Section 7 discusses limitations, and Section 8 concludes.

2. Related works

2.1. LLM Agents in Financial Analysis

The application of LLMs to financial tasks has grown rapidly, spanning earnings call analysis, financial question answering, and report summarisation. FinGPT [5] and BloombergGPT [6] demonstrate that domain-adapted language models outperform general-purpose ones on financial benchmarks. More relevant to our work, TradingAgent [7] and QuantAgent frame trading as a multi-step reasoning problem, delegating sub-tasks-market perception, reflection, and decision-to coordinated LLM agents. Our framework builds directly on this paradigm but extends it with vision-language chart analysis, quantitative alpha integration, and emerging-market adaptations absent from prior systems.

2.2. Technical Analysis and Pattern Recognition

Automating technical analysis has been explored through convolutional neural networks for candlestick pattern recognition [8], reinforcement learning for indicator-based policy learning [9], and hybrid rule-based systems [10]. Recent work has begun replacing handcrafted features with multi-modal chart-understanding models. ChartLlama [11] shows that vision-language models can extract actionable insights from charts without explicit feature engineering. Our system employs this capability within two dedicated vision agents, integrating it into a broader multi-agent decision pipeline rather than treating it in isolation.

2.3. Quantitative Alpha Factor Research

Systematic factor construction has a long tradition in quantitative finance, from foundational momentum and value factors [12, 13] to the large-scale formulaic approach of Kakushadze, who codified 101 analytic alpha expressions (WorldQuant 101). Subsequent work adapts these formulas to

single-asset settings [14] and evaluates their decay and capacity in different market regimes [15, 16]. However, traditional factor mining is notoriously susceptible to data-snooping and selection bias, pitfalls extensively documented by Harvey, Liu, and others. While modern non-LLM multimodal pipelines attempt to mitigate this by fusing factors with alternative data via standard machine learning architectures, they often lack the explicit semantic reasoning needed to contextualize anomalies. We extend this literature by constructing an 85-candidate registry-combining the full WorldQuant 101 corpus adapted to single-stock OHLCV with five proprietary formulas-and selecting the top-five factors per symbol dynamically via walk-forward Information Coefficient and Sharpe ratio ranking.

2.4. Sentiment Analysis in Finance and Vietnamese NLP

News sentiment has been established as a statistically significant predictor of short-term price movements [17], with transformer-based models such as FinBERT [18] setting the standard for English financial text. For Vietnamese, ViSoBERT provides a BERT-based model pre-trained on large-scale Vietnamese corpora and achieves strong results on Vietnamese sentiment classification benchmarks. Financial-domain validity does not follow automatically from general-language benchmark performance. We therefore configure a hosted ViSoBERT endpoint as the primary scorer while retaining an explicit lexicon fallback, and record the actual scoring path for every newly scored article. Target and related-company sentiment are presented in a separate LLM text context and do not enter the OHLCV factor formulas or ranking. Legacy cache entries that lack scorer metadata are treated as unknown rather than retrospectively attributed to ViSoBERT.

2.5. Trading Systems for Emerging Markets

Research on algorithmic trading in emerging markets is comparatively sparse. Studies on the Vietnamese market have primarily examined market efficiency [19], herding behaviour [20], and the predictive power of classical technical indicators under the price-limit regime [21], but none employs LLM-based agents. Our work develops a multi-agent LLM forecasting framework that records Vietnamese cash-equity execution and settlement assumptions separately from the domestic news ecosystem and single-stock OHLCV data characteristics; we do not claim that a universal settlement rule fixes a model horizon.

2.6. Evaluation Standards and Agentic Benchmarks

Recent literature highlights the critical need for rigorous, standardized evaluation in LLM-driven trading. For instance, FinVision [22] shares our modular multi-agent approach and vision-based technical analysis but places much stronger emphasis on extensive ablations (e.g., of reflection modules) and comprehensive risk reporting (Sharpe, drawdowns). Furthermore, an audit-oriented survey on financial LLMs [23] identifies widespread methodological gaps in time-consistent splits, transaction-cost modeling, and reproducibility. While our framework addresses costs and lookahead leakage, we recognize that our reporting detail currently falls short of the rigorous bar for full reproducibility. Finally, live/prediction-market benchmarks [24, 25] and forward-looking evaluation paradigms like FutureX [26] underscore that static walk-forward backtests—while useful—must ultimately be complemented by live trials and execution realism to substantiate strong claims about agentic efficacy.

3. Problem Formulation

3.1. Market Model and Data

Let $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ denote the universe of Vietnamese equities listed on HOSE, HNX, or UPCoM. For each asset $s \in \mathcal{S}$, the market is observed as a discrete-time OHLCV series:

$$\mathbf{x}_u = (O_u, H_u, L_u, C_u, V_u) \in \mathbb{R}_+^5, \quad u = 0, 1, \dots, T-1, \quad (1)$$

Algorithm 1: Walk-Forward Evaluation

Input: OHLCV series $\mathbf{X}$, alpha registry $\mathcal{A}$, historical sentiment cache $\mathcal{D}$, window size W , lookahead L , step size S

Output: Directional predictions $\hat{y}_{e,L}$, accuracy, Alpha Lift, and compounded account return

```
1 Initialize results list  $\mathcal{R} \leftarrow \emptyset$ 
2 foreach test point  $e$  in the walk-forward schedule do
3   Extract the exact analysis window  $\mathbf{X}_e = \mathbf{X}_{[e-W, e)}$ 
4   Compute technical indicators using TA-Lib
5   Evaluate all candidate alphas  $\alpha^{(k)} \in \mathcal{A}$ 
6   Remove candidates with no fitted transform, fewer than 20 finite evaluation observations, or
   non-finite ranking metrics to form  $\mathcal{V}$ 
7   Min-max normalise each metric over  $\mathcal{V}$  and rank using:
8
           
$$S_k = 0.40\mathcal{N}_{\mathcal{V}}(|IC_k|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc_k) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe_k)$$

9   Select top-5 alpha factors  $\mathcal{A}^*$ 
10  Retrieve historical CafeF sentiment at the decision cutoff  $d = e - 1$ 
11  Generate candlestick and trendline chart images from the same exact  $\mathbf{X}_e$  window
12  Run the Indicator, Alpha, Pattern, and Trend Agents
13  Pass all reports to Decision Agent
14  Obtain binary prediction  $\hat{y}_{e,L} \in \{\text{LONG}, \text{SHORT}\}$ 
15  Compare prediction with realised return
16  Store metrics in  $\mathcal{R}$ 
17 Compute directional accuracy and Alpha Lift
18 Compute compounded account return under Eq. (34)
19 return  $\mathcal{R}$ 
```

where O_u, H_u, L_u, C_u, V_u denote the open, high, low, closing price, and trading volume at zero-based bar position u respectively. When volume observations are missing for more than 50% of bars, the intra-bar range $H_u - L_u$ is substituted as a liquidity proxy.

We use zero-based row positions and an exclusive window-end boundary. For a test point with boundary e , the analysis window of size W is exactly:

$$\mathbf{X}_e = (\mathbf{x}_{e-W}, \mathbf{x}_{e-W+1}, \dots, \mathbf{x}_{e-1}) \in \mathbb{R}^{W \times 5}. \quad (\text{equivalently, } \mathbf{X}_e = \mathbf{X}_{[e-W, e)}) \quad (2)$$

Thus e is not itself a visible bar: the final visible (decision) bar is $d = e - 1$. The next bar, e , is reserved for the earliest simulated entry, and the immutable label target is $h = e - 1 + L$. A candidate economic exit may use that close only if the separate execution contract accepts it. This mapping is used in the engine, alpha labels, persisted provenance, and the experiments:

$$\boxed{\text{window} = [e - W, e), \quad d = e - 1, \quad \text{entry} = e, \quad h = e - 1 + L.} \quad (3)$$

The mapping is positional rather than calendar arithmetic; timestamps are carried alongside every index, so a weekend or holiday gap does not create a synthetic bar. The first complete window is $e = W$, and it is evaluable only when the target is present, i.e., $T \geq W + L$. For $L = 1$, the target is the entry bar ($h = e$), while for $L = 3$ it is two positions after entry ($h = e + 2$). Any boundary whose target is outside the available frame is rejected rather than shortened.

3.2. Directional Prediction Task

The core task predicts whether a candidate LONG cycle is profitable after the configured transaction costs. With $d = e - 1$ and $h = e - 1 + L$, define

$$r_{e,L}^{\text{net}} = \frac{C_h (1 - c_{\text{slip}})(1 - c_{\text{fee}})}{O_e (1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \quad y_{e,L} = \begin{cases} +1, & r_{e,L}^{\text{net}} > 0, \\ -1, & r_{e,L}^{\text{net}} \leq 0. \end{cases} \quad (4)$$

Here O_e is the next-bar entry open and C_h is the target close. The same net Open-to-Close return defines the directional label, the alpha-selection target, and the realised return of an executed LONG. Thus an opening gap and both transaction legs are represented consistently. The reported study uses daily bars with $L = 3$; intraday execution is outside its empirical scope.

A trading system $\mathcal{F}$ maps each analysis window to a binary decision:

$$\hat{y}_{e,L} = \mathcal{F}(\mathbf{X}_e) \in \{-1, +1\}, \quad (5)$$

where $+1$ corresponds to a **LONG** signal and -1 to **SHORT**.

3.3. Alpha Factor Formalisation and Adaptation

An *alpha factor* $\alpha^{(k)}$ is a measurable function that maps a feature frame derived from $\mathbf{X}_e$ to a scalar signal:

$$\alpha^{(k)} : \mathbb{R}^{T_{\text{hist}} \times d} \longrightarrow \mathbb{R}, \quad (6)$$

where d is the number of engineered features and T_{hist} is the historical window used for computation.

While the system incorporates the WorldQuant-101 corpus, these formulas were originally designed for cross-sectional ranking across a broad, highly liquid equity universe. To apply them to the Vietnamese market's single-stock setting, we introduce a *semantic translation layer* that redefines cross-sectional operators into rolling time-series equivalents.

Operator Localisation.. The primary transformation involves the cross-sectional $\text{rank}(x)$ operator, which denotes the relative percentile of a stock against its peers. We adapt this to a time-series percentile rank $\text{ts_rank}(x_u, w)$, representing the position of the current value x_u relative to its own history over a rolling window w :

$$\text{ts_rank}(x_u, w) = \frac{1}{w-1} \sum_{i=1}^{w-1} \mathbf{1}[x_u > x_{u-i}]. \quad (7)$$

The window parameter w is inherited from the original WQ-101 specification and is typically between 5 and 20 bars, depending on the formula. It is kept unchanged to preserve each formula's intended look-back horizon. Operators that reference cross-sectional averages or market returns are re-parametrised using the asset's own rolling history: Average Daily Volume (ADV_n) uses a n -day rolling mean of the asset's volume ($n = 10$ or 20 as specified in the original formula), and the Volume Weighted Average Price (VWAP) is approximated by the daily intra-bar proxy $(H_u + L_u + C_u)/3$. Time-series functions such as $\delta(x, d)$ and $\text{delay}(x, d)$ are implemented as d -period first differences and lags respectively, with d taken directly from the original formula. The minimum look-back required to compute a candidate is at least $\max(w) + d$ valid observations; candidates that cannot be evaluated within the 60-candle minimum history are excluded from selection at that test point.

Consistent Normalisation.. To ensure comparability across candidates with different scales, each adapted raw alpha value is normalised to $[-1, 1]$ via:

$$\tilde{\alpha}_u^{(k)} = \tanh\left(\frac{\alpha_u^{(k)} - \mu_k}{\sigma_k + \varepsilon}\right), \quad (8)$$

where μ_k and σ_k are the rolling mean and standard deviation of $\alpha^{(k)}$ over the calibration history, and $\varepsilon > 0$ prevents division by zero. The resulting signal $\tilde{\alpha}_u^{(k)} \in (-1, 1)$ is treated as a continuous directional score: positive values favour **LONG** and negative values favour **SHORT**.

3.4. Walk-Forward Alpha Selection

Let $\mathcal{A} = \{\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(M)}\}$ denote the registry of $M = 85$ candidate alpha functions. Each point-in-time history ends at the decision cutoff and is therefore disjoint from the current entry and target bars. Given the net Open-to-Close execution-return series $r_u^{\text{net},(L)}$ defined as in Eq. (4), the following diagnostics are computed on finite historical observations:

Information Coefficient (IC).. Following Grinold and Kahn [27], the Spearman rank correlation between the normalised alpha signal and forward returns over the evaluation suffix:

$$\text{IC}_k = \text{SpearmanCorr}(\tilde{\alpha}_{\mathcal{T}_{\text{eval}}}^{(k)}, r_{\mathcal{T}_{\text{eval}}}^{\text{net},(L)}). \quad (9)$$

Directional Accuracy.. The fraction of bars where the predicted direction agrees with the realised direction, restricted to bars with a strong enough signal:

$$\text{Acc}_k = \frac{1}{|\mathcal{T}^*|} \sum_{u \in \mathcal{T}^*} \mathbf{1}[\text{sign}(\tilde{\alpha}_u^{(k)}) = y_u^{(L)}], \quad (10)$$

where $\mathcal{T}^* = \{u : |\tilde{\alpha}_u^{(k)}| > 0.05\}$ excludes near-zero signals and $y_u^{(L)}$ is the canonical binary label from Eq. (4); hence a zero net return is **DOWN**, not a third class.

Long-Only Win Rate.. The precision of the alpha on bullish signals:

$$\text{LongAcc}_k = \frac{\sum_u \mathbf{1}[\tilde{\alpha}_u^{(k)} > 0.05 \wedge r_u^{\text{net},(L)} > 0]}{\sum_u \mathbf{1}[\tilde{\alpha}_u^{(k)} > 0.05]}. \quad (11)$$

Horizon Sharpe diagnostic.. For ranking, define $z_u^{(k)} = \text{sign}(\tilde{\alpha}_u^{(k)})r_u^{\text{net},(L)}$ on the finite historical sample. Following Sharpe [28], the diagnostic is:

$$\text{Sharpe}_k = \sqrt{\frac{252}{L}} \cdot \frac{\mathbb{E}_{u \in \mathcal{T}_{\text{eval}}} [z_u^{(k)}]}{\text{Std}_{u \in \mathcal{T}_{\text{eval}}} [z_u^{(k)}] + \varepsilon}. \quad (12)$$

Sortino, maximum drawdown, hit rate, long-only accuracy, and average trade are retained as descriptive candidate diagnostics but do not enter the composite selection score.

Candidates with no fitted transform, fewer than 20 finite evaluation observations, or a non-finite value for any ranking metric are excluded before scaling. Let $\mathcal{V} \subseteq \mathcal{A}$ be the remaining eligible set and let $q \in \{|\text{IC}|, \text{Acc}, \text{Sharpe}\}$. The implementation applies candidate-wise min-max scaling:

$$\mathcal{N}_{\mathcal{V}}(q_k) = \begin{cases} \frac{q_k - q_{\min}}{q_{\max} - q_{\min}}, & q_{\max} - q_{\min} \geq \varepsilon_{\text{rank}}, \\ \frac{1}{2}, & \text{otherwise,} \end{cases} \quad (13)$$

where $q_{\min} = \min_{j \in \mathcal{V}} q_j$, $q_{\max} = \max_{j \in \mathcal{V}} q_j$, and $\varepsilon_{\text{rank}} = 10^{-9}$. The default composite selection score is therefore:

$$S_k = 0.40 \mathcal{N}_{\mathcal{V}}(|\text{IC}_k|) + 0.35 \mathcal{N}_{\mathcal{V}}(\text{Acc}_k) + 0.25 \mathcal{N}_{\mathcal{V}}(\text{Sharpe}_k). \quad (14)$$

The weights prioritise association and directional accuracy while retaining a risk-adjusted return diagnostic. Long-only accuracy is reported but excluded from ranking to avoid favouring factors solely because the sample trends up.

Signal orientation is inferred from the sign of IC on the cutoff-safe historical snapshot. A negative-IC candidate is multiplied by -1 for its historical directional metrics and for the current inference value. All labels used for this operation have matured by the decision cutoff, but estimating orientation and ranking on the same finite history remains a selection risk discussed in Section 7.

Let $K = \min(5, |\mathcal{V}|)$. The top candidates are selected as:

$$\mathcal{A}^* = \arg \max_{\mathcal{B} \subseteq \mathcal{V}, |\mathcal{B}|=K} \sum_{\alpha^{(k)} \in \mathcal{B}} S_k. \quad (15)$$

3.5. Sentiment Signal Construction

Let $\mathcal{D}_e = \{d_1, d_2, \dots, d_{|\mathcal{D}_e|}\}$ be the set of CafeF articles available to the system for asset s at the timestamp τ_d of decision bar $d = e - 1$. Each article is assigned a score by the scorer actually recorded for that article:

$$\rho_i = \text{Scorer}_i(d_i) \in [-1, 1], \quad (16)$$

where $\rho_i = +1$ denotes strongly positive, -1 strongly negative, and 0 neutral sentiment. For newly scored records, Scorer_i is either the configured **5CD-AI/Vietnamese-Sentiment-visobert** HF Inference API endpoint or the versioned deterministic lexicon fallback. Each record stores the scorer, requested model identifier, observed Hub revision and its verification status, and fallback status/reason.

Legacy scores without these fields are marked **legacy-unknown**; no model identity is inferred from their numeric values.

The unweighted mean sentiment score for descriptive reporting is:

$$\bar{\rho}_e = \frac{1}{|\mathcal{D}_e|} \sum_{i=1}^{|\mathcal{D}_e|} \rho_i. \quad (17)$$

For the causal sentiment context, each dated article receives an exponential weight based on its age at cutoff τ_d :

$$\Delta_i(\tau_d) = \tau_d - \text{date}(d_i), \quad (18)$$

$$w_i(\tau_d) = 2^{-\Delta_i(\tau_d)/h}, \quad h = 30 \text{ days}, \quad (19)$$

$$S_{\text{decay},e} = \frac{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d) \rho_i}{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)}, \quad (20)$$

$$\text{Rel}_{\text{sent},e} = S_{\text{decay},e} - \frac{1}{|\mathcal{R}_e|} \sum_{s' \in \mathcal{R}_e} S_{\text{decay},e}^{(s')}, \quad (21)$$

where $\mathcal{R}_e$ contains only related companies with at least three valid dated-and-scored articles at cutoff τ_d . The exponential mean makes recent articles more influential without comparing a batch mean against the same batch from which it was calculated. $\text{Rel}_{\text{sent},e}$ compares like-for-like time-decayed scores.

Batch definition and low-news stability.. The set $\mathcal{D}_e$ contains all dated articles for target stock s in the 90-day sliding window ending at cutoff τ_d . The complete compact observation set is used for Eq. 20; the 15-article list stored for display is not the normalization sample. With fewer than three valid dated-and-scored observations, $S_{\text{decay},e}$ is marked unavailable rather than assigned a neutral zero. Likewise, $\text{Rel}_{\text{sent},e}$ is unavailable when $\mathcal{R}_e$ is empty. The effective sample size, invalid-date/score counts, and number of post-cutoff observations excluded are stored in run provenance. Neither context variable enters any numerical alpha formula or the registry ranking.

To prevent look-ahead leakage during backtesting, every cached article stores timezone-aware `published_at` and `available_at` fields in `Asia/Ho_Chi_Minh`. The cache is keyed by (s, τ_d) and returns only articles satisfying `available_at`(d_i) $\leq \tau_d$. CafeF records having date-only precision are assigned a conservative availability time of 00:00 on the following local day; records without a parseable date are rejected in research mode and never used to fill a sparse historical window. All 180 articles in the nine supplied cache artifacts contain parseable dates, so the previous undated fallback was a latent risk rather than evidence that those reported runs actually consumed undated articles.

3.6. Multi-Agent Decision Function

The full system decomposes $\mathcal{F}$ into five sub-functions corresponding to the five specialised agents:

$$\mathbf{r}^{\text{ind}} = \mathcal{F}_{\text{ind}}(\mathbf{X}_e), \quad (22)$$

$$\mathbf{a}_e = \mathcal{F}_{\text{factor}}(\mathbf{X}_e, \mathcal{A}^*), \quad (23)$$

$$\mathbf{c}_e^{\text{sent}} = \mathcal{F}_{\text{context}}(\mathcal{D}_e, S_{\text{decay},e}, \text{Rel}_{\text{sent},e}), \quad (24)$$

$$\mathbf{r}^{\text{alpha}} = \mathcal{F}_{\text{text}}(\mathbf{a}_e, \mathbf{c}_e^{\text{sent}}), \quad (25)$$

$$\mathbf{r}^{\text{pat}} = \mathcal{F}_{\text{pat}}(\mathbf{I}_e^{\text{kline}}), \quad (26)$$

$$\mathbf{r}^{\text{trend}} = \mathcal{F}_{\text{trend}}(\mathbf{I}_e^{\text{trend}}), \quad (27)$$

where $\mathbf{I}_e^{\text{kline}}$ and $\mathbf{I}_e^{\text{trend}}$ are the candlestick and trendline-annotated chart images rendered from $\mathbf{X}_e$, and $\mathbf{c}_e^{\text{sent}}$ is a separate textual context containing the target and related-company sentiment summaries. The factor values in $\mathbf{a}_e$ are functions of $\mathbf{X}_e$ only; the LLM compares them qualitatively with $\mathbf{c}_e^{\text{sent}}$ when producing $\mathbf{r}^{\text{alpha}}$.

The Decision Agent aggregates all four reports through structured chain-of-thought prompting [29] and produces the final binary signal:

$$\hat{y}_{e,L} = \mathcal{F}_{\text{dec}}(\mathbf{r}^{\text{ind}}, \mathbf{r}^{\text{alpha}}, \mathbf{r}^{\text{pat}}, \mathbf{r}^{\text{trend}}) \in \{-1, +1\}. \quad (28)$$

The forced binary output constraint ensures $\hat{y}_{e,L} \neq 0$ for every valid *prediction*; it does not imply an always-invested account. Under the default long-only action map, **SHORT** becomes **CASH**. **LONG** executes a complete next-open-to-target-close cycle when valid positive prices and timestamps are available. This separation permits binary directional evaluation without misrepresenting a bearish prediction as an executed short sale.

3.7. Walk-Forward Evaluation Protocol

We adopt a rolling walk-forward evaluation protocol commonly used in quantitative trading system evaluation [30].

Given a historical series of T bars, valid test points are exclusive boundaries with $W \leq e \leq T - L$. The implementation uses the latest available boundaries at spacing S (up to N_{test} points) and processes them in chronological order:

$$\mathcal{T} = \{e_i = T - L - (i - 1) \cdot S \mid i = 1, 2, \dots, N_{\text{test}}\}, \quad (29)$$

retaining only values $e_i \geq W$ before reversing the list for execution. Thus $e = W$ is included whenever it is among the requested points, and every selected point satisfies the complete-window and complete-target conditions. Here S is the step size between consecutive test points and $N_{\text{test}} = 20$ in our main experiments.

The three primary evaluation metrics are:

$$\text{Acc} = \frac{1}{|\mathcal{T}|} \sum_{e \in \mathcal{T}} \mathbf{1}[\hat{y}_{e,L} = y_{e,L}], \quad (30)$$

$$\Delta\text{Acc} = \text{Acc}_{\text{Full}} - \text{Acc}_{\text{No-}\alpha}, \quad (31)$$

$$q_e = a_e \left[\frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1 \right], \quad (32)$$

$$W_{k+1} = W_k (1 + q_{e_k}), \quad W_0 = 1, \quad (33)$$

$$R_{\text{acct}} = \frac{W_K}{W_0} - 1, \quad (34)$$

where ΔAcc (*Alpha Lift*) quantifies the improvement in directional accuracy attributable to the Alpha Agent. In the economic simulation, $a_e = 1$ for **LONG** and $a_e = 0$ for **SHORT**. Each **LONG** allocates all available cash at O_e , pays fee and slippage on the BUY leg, liquidates all shares at C_{e-1+L} , and pays the same rates on the SELL leg. The configured values are $c_{\text{fee}} = 0.25\%$ and $c_{\text{slip}} = 0.10\%$ per leg. Each test point is a closed cycle and the implementation requires $S \geq L$, so no shares carry into the next point. **SHORT** forecasts hold cash and incur neither fee nor slippage.

Let $\rho_k = W_{k+1}/W_k - 1 = q_{e_k}$ denote the cycle return, including zeros for CASH. The account risk metrics are

$$\text{Sharpe}_{\text{acct}} = \sqrt{252} \frac{\mathbb{E}[\rho_k]}{\text{Std}[\rho_k] + \varepsilon}, \quad (35)$$

$$\text{MDD}_{\text{acct}} = \max_{0 \leq k \leq K} \left(1 - \frac{W_k}{\max_{0 \leq i \leq k} W_i} \right). \quad (36)$$

The Sharpe risk-free rate is zero, and the population standard deviation is used. Maximum drawdown is a relative wealth loss and includes the initial wealth anchor W_0 , so an initial losing cycle is not omitted.

Finally, let $\mathcal{X} = \{e_k : a_{e_k} = 1\}$ be the executed-position sample. Trade-level summaries are

$$\text{HitRate} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} \mathbf{1}[q_e > 0], \quad \text{AvgTrade} = \frac{1}{|\mathcal{X}|} \sum_{e \in \mathcal{X}} q_e, \quad (37)$$

with both defined as zero when $\mathcal{X}$ is empty. SHORT/CASH cycles do not dilute either executed-trade denominator. Risk ratios are reported as zero when their dispersion denominator does not exceed the numerical tolerance. The cost rates are research allowances rather than a claim about a particular venue’s realised fee schedule.

4. System architecture

4.1. Overview

The implemented runtime is a LangGraph directed acyclic graph with five specialised agents and a shared typed state object. Its default configuration names `openai/gpt-oss-20b` for text and `qwen/qwen3.8-27b` for vision through Groq. The backtest path enforces point-in-time input windows, shared upstream reports for paired variants, structured binary decisions, and a separate compounded-account ledger.

4.2. Indicator Agent

The Indicator Agent computes five momentum and volatility oscillators using TA-Lib over the most recent OHLCV window, including MACD [31], RSI [32], Stochastic Oscillator [33], Williams %R [34], and ROC-based momentum signals [35]. Table 1 lists each indicator with its parameters and signal classification rules. Signal classification is performed deterministically in Python before any LLM call, using hard thresholds and two-bar crossover detection. The classified counts-number of bullish, bearish, and neutral signals-are injected directly into the LLM prompt as immutable ground-truth constraints, preventing the model from hallucinating signal tallies. A majority rule then determines an overall consensus with an associated confidence level (High if four or more signals agree, Medium if three agree, Low otherwise).

Table 1: Technical indicators and trading signal conditions

Indicator	Parameters	Bullish condition	Bearish condition
MACD	12, 26, 9	Golden cross or positive histogram expanding	Death cross or negative histogram expanding
RSI	period = 14	≤ 30 (oversold) or value ≥ 55	≥ 70 (overbought) or value ≤ 45
Stochastic	14, 3, 3	%K crosses above %D, or %K ≤ 20	%K crosses below %D, or %K ≥ 80
Williams %R	period = 14	%R ≤ -80 or trending toward 0	%R ≥ -20 or trending toward -100
ROC	period = 10	ROC $> +0.5\%$	ROC $< -0.5\%$

4.3. Alpha Agent

The Alpha Agent serves two explicitly separated functions: OHLCV-only quantitative factor evaluation and qualitative comparison with Vietnamese sentiment supplied as textual LLM context.

Alpha selection. An offline registry of 85 candidate alpha functions is maintained, comprising five proprietary single-stock momentum and reversion formulas (Table 2) and 80 adaptations of the WorldQuant 101 Formulaic Alphas to single-stock OHLCV data. Cross-sectional operators such as `ts_rank` are adapted to their single-asset time-series analogues (i.e., rolling percentile rank over the lookback window). We acknowledge that this translation alters the statistical properties of the original factors: without the cross-sectional dispersion that drives many WQ-101 alphas, single-asset surrogates may exhibit faster signal decay, higher autocorrelation, and reduced distributional stability. A full per-factor diagnostic examining IC decay profiles, signal autocorrelation, and factor turnover rates is left to future work (see Section 7). When the volume series is unavailable or sparse, intra-bar price range is used as a proxy.

For a given symbol and timeframe, the available point-in-time history is truncated to at most 600 bars ending at the decision cutoff. Features (returns, log-volume, VWAP proxy, and ADV20) are then constructed. Candidate metrics use only matured net Open-to-Close labels within this snapshot; the final L rows are excluded because their targets are unavailable at the cutoff. A zero net return is assigned `DOWN`, consistent with the requirement that a `LONG` must remain profitable after costs.

The selector computes Spearman Information Coefficient (IC), directional accuracy on signals above the configured absolute threshold, and a horizon Sharpe diagnostic. Long-only win rate and other economic summaries are retained for diagnosis but do not enter the composite score.

A candidate is eligible only when it has at least 20 finite observations and finite ranking metrics. Ineligible candidates are removed *before* the per-metric min-max bounds are computed. Using $\mathcal{N}_V$ from Equation (13), the default composite score is:

$$S = 0.40\mathcal{N}_V(|IC|) + 0.35\mathcal{N}_V(Acc) + 0.25\mathcal{N}_V(Sharpe) \quad (38)$$

If a metric is constant across the eligible set, its normalised value is 0.5 for every candidate. The displayed coefficients are the default runtime weights; robustness configurations replace them with recorded alternatives. Long-only accuracy is excluded from ranking to avoid rewarding a factor solely because the historical sample trends upward.

Up to five eligible candidates are selected for the current prediction. When historical IC is negative, the signal is multiplied by -1 for metric computation and current inference. All observations

used for this orientation precede the decision cutoff, but estimating orientation and rank on the same finite history can still overfit to noise. Bootstrapped IC confidence intervals or a Deflated Sharpe adjustment [36, 15] would provide stronger guarantees of signal stability.

Table 2: Five proprietary alpha formulas

Alpha	Name	Formula
α_1	Flow-Driven Momentum	$\tanh(\text{ROC}(1)_{\text{adj}} \times \text{VolSurge} \times 0.5)$
α_2	MACD-Flow Proxy	$\tanh(\text{MACDHist}_{\text{norm}} + \text{ROC}(1)_{\text{adj}} \times 0.5)$
α_3	Liquidity Void Reversion	$\tanh\left(-0.8 \times \text{ZScore}\left(\frac{\text{Close} - \text{SMA}_5}{\text{Close}}\right)\right)$
α_4	Bollinger Squeeze & Flow	$\tanh((\%B - 0.5) \times \text{VolZ}_5 \times 1.5)$
α_5	Order Flow Exhaustion	$\tanh(\text{ROC}(1)_{\text{adj}} \times \text{ClosePos} \times 3.0)$

Sentiment integration.. Articles from CafeF are crawled for the target ticker and up to ten related companies identified by the LLM. The primary scoring path calls the configured **5CD-AI/Vietnamese-Sentiment-visobert** HF Inference API endpoint; endpoint failure triggers a versioned deterministic lexicon scorer. Every newly scored article stores the actual scorer, requested model identifier, observed Hub revision/status, and fallback status/reason. A run reports the ViSoBERT endpoint identifier only when all included articles verify successful use of that endpoint; mixed, fallback-only, and legacy-unknown runs are labelled accordingly.

Two point-in-time context variables are constructed. S_{decay} is the exponentially time-weighted mean of all valid dated-and-scored articles in the 90-day window, with a 30-day half-life and strict exclusion of post-cutoff observations. Rel_{sent} is the difference between the target’s time-decayed score and the mean time-decayed score of related companies having at least three valid observations. Either value is marked unavailable when its observation requirement is not met; missing data is not encoded as neutral zero. The full compact observation set is used for these calculations, while the 15-article slice is display-only.

Neither variable is injected into α_2 , any other registry formula, or factor ranking. They and the per-company related sentiment values are rendered in a separate prompt section that the LLM uses only for qualitative comparison with the OHLCV factor report.

In backtest mode, the live crawler is replaced by a pre-loaded historical sentiment store keyed by symbol and decision timestamp. Inclusion is based on timezone-aware **available_at**, not merely a publication date: date-only CafeF records become available at 00:00 on the next day in **Asia/Ho_Chi_Minh**, and undated records are rejected rather than used as fallback.

4.4. Pattern Agent

The Pattern Agent analyses a candlestick chart rendered from the current analysis window $[e - W, e)$ (45 candles in the main configuration). The chart is produced by **mplfinance** using a custom colour scheme (green bodies for bearish candles and red bodies for bullish candles, following Vietnamese market convention) and encoded as a base64 PNG before graph execution.

The vision-language model is prompted to perform four sequential tasks. First, it identifies the dominant chart formation from a curated list of 16 classical patterns, such as head-and-shoulders inverse, double bottom, or various triangles and wedges. Second, it describes the three to five most recent candles in terms of body size, wick length, and relative position. Third, it assesses whether a

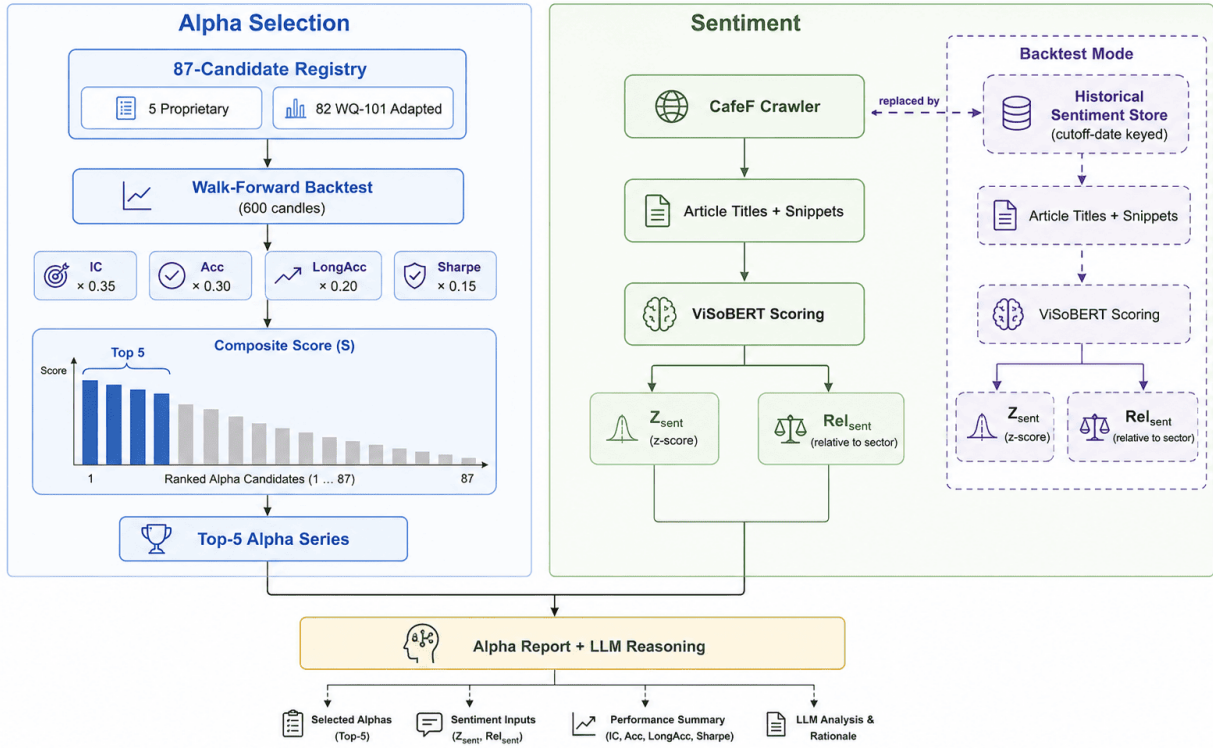


Figure 1: Internal data flow of the Alpha Agent

breakout or breakdown is occurring. Finally, it commits to a directional bias (**bullish**, **bearish**, or **neutral**) alongside a confidence rating of **High**, **Medium**, or **Low**.

The prompt is explicitly structured to discourage non-committal answers and encourage a directional interpretation whenever sufficient evidence is present.

After inference, a deterministic verifier compares the reported bias with the final candlestick body and the direction of the last five closes. An obvious contradiction is appended to the report and downgrades its confidence. If the vision model fails or no chart image is available, the agent falls back to a text-only analysis of raw OHLCV values using the text LLM.

4.5. Trend Agent

The Trend Agent receives a separate chart in which support and resistance lines have been fitted algorithmically before LLM invocation.

Two trendline pairs are computed.

Close-price trend channel. The first pair is derived from the closing price series alone. A linear regression line is fitted to all 45 candles in $[e - W, e)$, and the upper and lower deviations from this line identify the resistance and support pivots respectively. The slopes are then refined by minimising squared deviations subject to two constraints: the support line must not intersect any closing price from above, and the resistance line must not intersect any closing price from below.

High-low channel. The second pair uses the high and low price series to identify the highest and lowest deviation points, producing channel lines that bound the full intra-bar range.

The support lines are drawn in blue, resistance lines in red, and rendered at 600 DPI to preserve legibility when compressed to base64.

The vision-language model is prompted to assess the slope of each support and resistance line, evaluate the distance between the lines (indicating channel width and price compression), determine the position of the most recent candles relative to each boundary, and produce a directional forecast for the applicable prediction horizon. The required JSON output fields encompass the trend direction, numerical support and resistance levels, slope characterisation, and overall confidence.

After inference, a deterministic verifier checks the reported trend against price relative to SMA20, the availability of SMA50, and the recent price slope. Contradictory claims are flagged and their confidence is reduced. If the vision-language model fails or no rendered chart is available, the agent falls back to a text-only analysis of raw OHLCV data, mirroring the strategy used by the Pattern Agent.

4.6. Decision Agent

The Decision Agent synthesises all upstream reports into a single trading judgment. To remain within the Groq token-per-minute limit, each report is condensed before being passed to the prompt: the Indicator report is trimmed to its consensus summary block, the Alpha report retains only the factor table and LLM reasoning paragraph, and the Sentiment report drops the per-article listing.

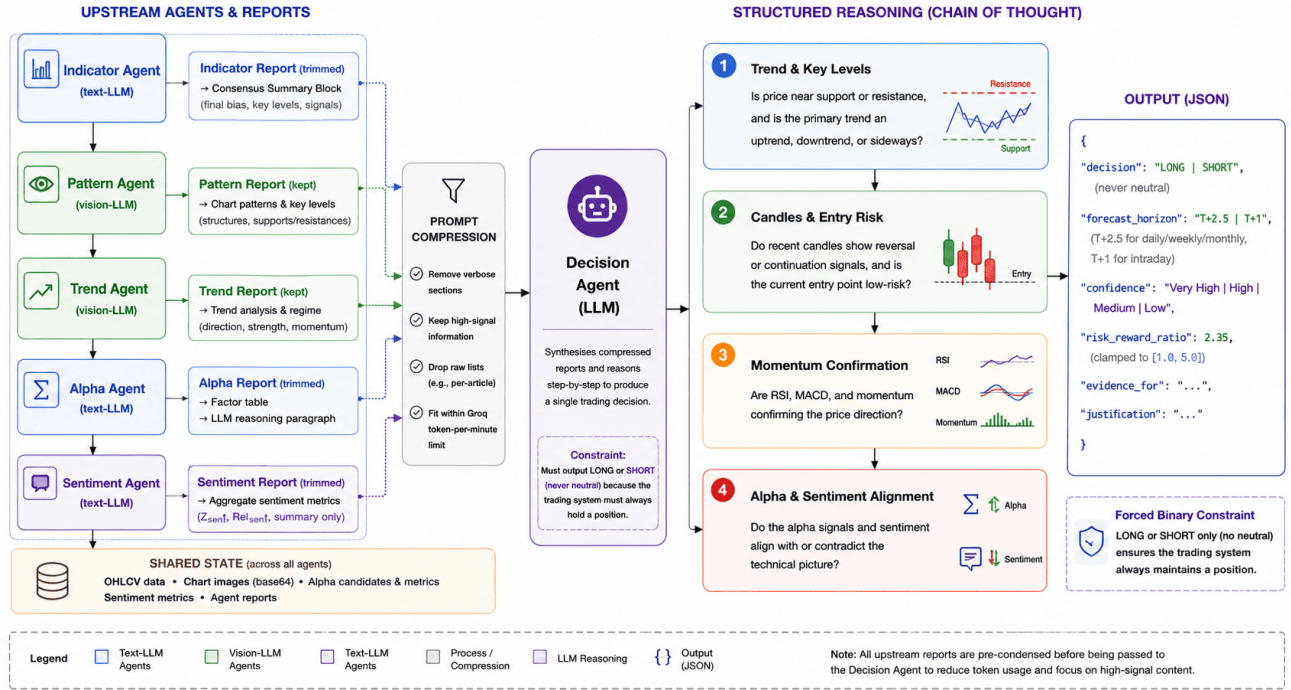


Figure 2: Decision Agent reasoning pipeline

The agent is prompted to reason through four sequential questions in a structured chain-of-thought process. It evaluates whether the price is near support or resistance while identifying the primary trend, checks if recent candles show reversal or continuation signals to ensure a low-risk entry, verifies whether technical oscillators like RSI and MACD confirm the price direction, and finally assesses if the quantitative alpha signals and sentiment align with or contradict the technical picture.

Following this reasoning, the model must output a JSON object comprising a forced binary direction forecast (LONG or SHORT), the `forecast_horizon`, the confidence level, a `risk_reward_ratio`

clamped to $[1.0, 5.0]$, supporting and opposing evidence, and a final `justification` paragraph. The prompt explicitly states that this label is not proof of an executed order.

The forced binary constraint supports direction classification only. Downstream account logic independently maps predictions to actions: `SHORT` maps to `CASH`, whereas `LONG` invests all available cash at the next open and closes the position at the configured target close after applying costs on both legs.

5. Experimental Setup

5.1. Dataset and Point-in-Time Data Contract

The daily evaluation universe contains nine Vietnamese equities spanning five economically distinct groups: information technology (FPT and CMG), banking (VCB and MBB), retail (MWG), food and beverages (VNM and BHN), and aviation and tourism (HVN and VJC). The purposive stratification combines the VN30/large-cap names specified in the frozen design (VCB, FPT, VNM, MBB, and MWG) with mid-cap or ownership-specialised firms (CMG, BHN, and HVN) and a large private airline (VJC). It was chosen to expose the agents to heterogeneous liquidity, ownership, and news environments; it is not claimed to be a random or market-representative sample. The benchmark contains 20 chronologically ordered test points per symbol, for 180 paired observations in total. Section 7 discusses the resulting external-validity limit, and Appendix Table E.12 reports the complete ticker and sentiment-coverage audit.

OHLCV observations are obtained through the project’s `vnstock`-based data loader. At an exclusive test boundary e , every agent sees only bars in $[e - W, e)$, so the last visible decision bar is $d = e - 1$. Dynamic alpha selection receives a cutoff-safe prefix ending at the same decision bar and containing at most 600 observations. Neither the factor calculations nor the chart renderers can inspect the entry bar or target bar. The numerical, candlestick, and trend inputs use the same requested window length and fail closed when a complete window is unavailable.

The historical sentiment store applies the same cutoff. Only records with a parseable publication date no later than d are eligible; undated records are rejected in research mode instead of being borrowed from a later cache state. Across the supplied caches, all 180 articles have a parseable date. The mean number available in a 90-day point-in-time window is 5.77, and 73 of 180 test points (40.56%) contain at least eight articles. The legacy cache schema does not retain the scorer identity or fallback flag, so 0 of 180 articles can be verified as ViSoBERT-scored, 0 can be verified as lexicon-scored, and all 180 are conservatively reported as `legacy-unknown`. These figures measure verifiable provenance rather than infer model identity from confidence or sentiment values. The sentiment pathway remains text context for the Decision Agent and does not enter any OHLCV alpha formula.

5.2. Walk-Forward Protocol

All reported experiments use daily observations. The main configuration is $W = 45$, step size $S = 3$, and lookahead $L = 3$. For boundary e , the simulated entry is the next open O_e and the target exit is C_{e-1+L} . The realised net return of a candidate LONG cycle is

$$r_{e,L}^{\text{net}} = \frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \quad (39)$$

where $c_{\text{fee}} = 0.0025$ and $c_{\text{slip}} = 0.001$ are applied separately on both the BUY and SELL legs. The economic direction label is `UP` only when $r_{e,L}^{\text{net}} > 0$ and is `DOWN` otherwise. This definition makes the classification target, alpha-selection target, and account return share the same Open-to-Close horizon and cost convention.

Table 3: Main walk-forward configuration.

Parameter	Value	Operational meaning
Analysis window W	45 bars	Exact agent input $[e - W, e)$
Step size S	3 bars	Separation between consecutive test boundaries
Lookahead L	3 bars	Entry at O_e and exit at C_{e-1+L}
Test points	20 per symbol	180 paired observations over nine symbols
Alpha history	Up to 600 bars	Cutoff-safe calibration/evaluation prefix ending at $d = e - 1$
Alpha registry	85 candidates	Top five selected at each test point
Random seeds	42	Python/NumPy and bootstrap reproducibility

5.3. Paired Baseline and Disentangled Ablations

The primary benchmark compares the Full System with a No-Alpha baseline under the implemented `paired_shared_reports` protocol. Indicator, Pattern, and Trend agents run once at each cutoff. Their immutable report snapshot is deep-copied into both decision branches, so only the Alpha bundle and the corresponding final decision differ. This eliminates upstream LLM and chart-generation variation within each pair. All 180 benchmark pairs produced valid binary decisions.

Table 4: Paired benchmark variants.

Component	Full System	No-Alpha baseline
Indicator report	Shared upstream snapshot	Identical shared snapshot
Pattern and Trend reports	Shared upstream snapshot	Identical shared snapshot
OHLCV alpha factors	Enabled	Disabled
Vietnamese sentiment context	Enabled	Disabled
Final Decision Agent	Receives all enabled reports	Receives only shared upstream reports

To separate the two components bundled in the primary intervention, a second experiment uses the `four_way_shared_upstream` protocol on FPT, VNM, and VCB. It evaluates Full (factors and sentiment), Alpha-only (factors), Sentiment-only (sentiment), and Baseline (neither). All four branches receive the same upstream snapshot at each test point. The independent directional contributions are defined relative to Baseline, while the interaction is

$$\text{Synergy} = \text{Acc}_{\text{Full}} - \max(\text{Acc}_{\text{AlphaOnly}}, \text{Acc}_{\text{SentimentOnly}}). \quad (40)$$

5.4. Account and Statistical Metrics

Each variant starts with VND 50,000,000 and is simulated independently. Every LONG uses all available cash at O_e , pays fee and slippage on purchase, sells all shares at C_{e-1+L} , and pays both costs again on sale. Every cycle therefore ends in cash before the next test point. A SHORT forecast is a directional prediction only and maps to CASH because uncovered short selling of the underlying equity is disabled. With cycle return $q_k = r_{e_k,L}^{\text{net}}$ for LONG and $q_k = 0$ for SHORT, wealth compounds as

$$W_{k+1} = W_k(1 + q_k), \quad R_{\text{acct}} = \frac{W_K}{W_0} - 1. \quad (41)$$

Account Sharpe is the mean of the 20 cycle returns divided by their population standard deviation and annualised by $\sqrt{252}$; CASH cycles contribute zero. Maximum drawdown is computed from the compounded cycle-end equity curve anchored at W_0 . These metrics describe the stated research simulation rather than a deployable execution strategy: spread variation, partial fills, taxes, liquidity limits, and market impact are outside the model.

Statistical inference is computed on paired common support. For each symbol, an exact two-sided McNemar test compares discordant correctness outcomes. A moving-block bootstrap with block length five, 1,000 replications, and seed 42 yields the 95% interval for Alpha Lift. A Newey–West test is also stored for the paired correctness-difference series. Across the nine symbols, a two-sided Wilcoxon signed-rank test evaluates whether the symbol-level lifts are centred at zero.

Table 5: Evaluation metrics and estimands.

Metric	Definition
Directional accuracy	Fraction of valid LONG/SHORT forecasts matching the net economic direction in Eq. (4).
Alpha Lift	Full accuracy minus the accuracy of its paired No-Alpha branch, in percentage points.
Account return	Compounded fixed-cycle return $W_K/W_0 - 1$; LONG executes and SHORT holds cash.
Account Sharpe	Mean cycle return divided by population standard deviation and annualised by $\sqrt{252}$.
Maximum drawdown	Largest peak-to-trough decline on the compounded cycle-end equity curve.
Paired inference	Exact McNemar test, moving-block 95% lift interval, Newey–West mean test, and cross-symbol Wilcoxon signed-rank test.

5.5. Implementation and Robustness Design

The text model is `openai/gpt-oss-20b` and the vision model is `qwen/qwen3.8-27b`; both use temperature zero in the reported runs. Hosted inference can nevertheless remain nondeterministic. Calls are executed sequentially with rate-limit pacing, and checkpoint files permit completed test points to be resumed without changing their stored outputs.

Table 6: Implementation configuration used by the reported experiments.

Component	Configuration
Text / vision models	openai/gpt-oss-20b / qwen/qwen3.8-27b; temperature 0.0
Initial capital	VND 50,000,000 independently for each variant
Costs	0.25% fee and 0.10% slippage on each BUY and SELL leg
Position rule	LONG invests all cash for one fixed cycle; SHORT remains in CASH
Primary protocol	paired_shared_reports; 20 test points per symbol
Four-way protocol	four_way_shared_upstream; Full, Alpha-only, Sentiment-only, Baseline

The robustness experiment evaluates FPT and VNM with 20 test points per configuration. Panel A changes the common analysis window to $W \in \{30, 45, 60\}$. Panel B compares `zscore_tanh`, `minmax`, and `rank` alpha normalisation. Panel C compares default, equal, and IC-heavy ranking weights. For Panels B and C, the No-Alpha branch is reused from the same shared upstream/control checkpoint within each symbol and panel; its decisions must therefore be invariant to parameters that only affect the Alpha branch. Panel A deliberately changes the shared input window, so its No-Alpha output is not expected to remain constant across rows.

6. Backtest Results

6.1. Nine-Symbol Paired Benchmark

Table 7 reports the completed paired benchmark. The Full System attains mean directional accuracy of 65.0%, compared with 51.1% for No-Alpha, for a mean Alpha Lift of +13.9 percentage points (pp). Eight symbols have positive lift and VJC has zero lift. The cross-symbol Wilcoxon signed-rank test rejects a zero-centred lift ($p = 0.0078$), although none of the individual exact McNemar tests reaches the 5% level with only 20 paired points per symbol. The moving-block intervals are correspondingly wide; VCB is the only symbol whose reported 95% lift interval excludes zero.

The economic results are materially more cautious than the directional results. Full outperforms No-Alpha in account return for all nine symbols, but its mean account return is still -1.69% , and only HVN, MBB, and MWG finish positive. Hence the experiment supports a paired directional improvement and a relative reduction in losses, not general profitability. The divergence is expected under the long-only action map: a correct SHORT improves directional accuracy but earns a zero account return.

6.2. Robustness and Sensitivity

Table 8 reports all six completed sweeps. Each row contains 20 paired test points. Panel A changes the input window for both variants and therefore changes the No-Alpha information set. In Panels B and C, which intervene only on Alpha processing, the No-Alpha accuracy is exactly invariant within each symbol and panel: 45.0% for both normalisation sweeps, 50.0% for the FPT weight sweep, and 45.0% for the VNM weight sweep.

The alpha-enabled branch is sensitive to configuration. In particular, FPT at $W = 60$ produces a -5.0 pp lift, and VNM at $W = 30$ produces no lift. The normalisation and weighting panels remain positive for both symbols, but their Full accuracies and economic returns vary materially. These results validate the paired-control implementation while also showing that the intervention is not uniformly robust to window choice or alpha design.

Table 7: Paired benchmark over nine equities ($N = 20$ per symbol, $W = 45$, $S = 3$, $L = 3$). Panel A reports directional inference; Panel B reports compounded long-only account outcomes.

<i>Panel A: Directional accuracy and paired inference</i>						
Symbol	Full (%)	No- α (%)	Lift (pp)	95% CI (pp)	McNemar p	N
BHN	55.0	45.0	+10.0	[+0.0, +15.0]	0.6875	20
CMG	75.0	70.0	+5.0	[+0.0, +5.0]	1.0000	20
FPT	65.0	45.0	+20.0	[−5.0, +35.0]	0.2188	20
HVN	70.0	50.0	+20.0	[+0.0, +35.0]	0.2188	20
MBB	70.0	45.0	+25.0	[+0.0, +55.0]	0.1250	20
MWG	70.0	50.0	+20.0	[+0.0, +40.0]	0.1250	20
VCB	65.0	50.0	+15.0	[+5.0, +35.0]	0.2500	20
VJC	60.0	60.0	+0.0	[−10.0, +10.0]	1.0000	20
VNM	55.0	45.0	+10.0	[−15.0, +45.0]	0.6250	20
Mean	65.0	51.1	+13.9	—	—	20

<i>Panel B: Compounded account metrics</i>						
Symbol	Return F (%)	Return N (%)	Sharpe F	Sharpe N	MDD F (%)	MDD N (%)
BHN	-4.20	-5.76	-3.63	-2.64	4.80	8.31
CMG	-2.66	-5.60	-3.64	-5.28	2.66	5.60
FPT	-1.56	-10.37	-0.86	-6.41	6.30	10.37
HVN	+2.89	-15.64	+1.19	-4.27	5.27	20.06
MBB	+5.21	-7.78	+2.36	-4.22	3.50	9.89
MWG	+0.39	-6.42	+0.41	-3.93	1.92	8.17
VCB	-8.34	-10.49	-5.98	-7.78	8.92	10.49
VJC	-6.05	-6.26	-5.22	-5.38	6.05	6.26
VNM	-0.85	-7.51	-0.41	-4.91	3.40	7.51
Mean	-1.69	-8.43	-1.75	-4.98	4.76	9.63

F and N denote Full and No-Alpha. The 95% intervals use a moving-block bootstrap (block length 5; 1,000 replications). The two-sided cross-symbol Wilcoxon result is $p = 0.0078125$ (nine symbol-level lifts; median lift = 15.0 pp). Returns are compounded from VND 50,000,000 independent accounts under Eq. (34); SHORT forecasts remain in cash.

6.3. Disentangled Alpha–Sentiment Ablation

Table 9 separates quantitative factors from Vietnamese sentiment context on three representative symbols. Alpha-only improves directional accuracy for every symbol, with a mean independent contribution of +8.33 pp. Sentiment-only contributes +5.0 pp for VNM and 0.0 pp for FPT and VCB, giving a mean independent contribution of +1.67 pp. The interaction is heterogeneous: +5.0 pp for FPT, +10.0 pp for VNM, and −10.0 pp for VCB.

The ablation does not support attributing the entire primary benchmark lift to sentiment. The

Table 8: Robustness sweep on FPT and VNM ($N = 20$ per row). Returns are compounded account returns.

Symbol	Panel	Scenario	Full (%)	No- α (%)	Lift (pp)	Return F (%)	Return N (%)
FPT	Window	$W = 30$	55.0	40.0	+15.0	-2.63	-10.27
FPT	Window	$W = 45$	65.0	50.0	+15.0	-2.21	-6.86
FPT	Window	$W = 60$	40.0	45.0	-5.0	-8.04	-4.38
VNM	Window	$W = 30$	40.0	40.0	+0.0	-7.81	-7.51
VNM	Window	$W = 45$	55.0	40.0	+15.0	-3.59	-7.06
VNM	Window	$W = 60$	65.0	55.0	+10.0	+5.20	-0.84
FPT	Normalisation	zscore_tanh	50.0	45.0	+5.0	-4.72	-7.38
FPT	Normalisation	minmax	55.0	45.0	+10.0	-5.63	-7.38
FPT	Normalisation	rank	60.0	45.0	+15.0	-4.36	-7.38
VNM	Normalisation	zscore_tanh	60.0	45.0	+15.0	+2.70	-2.22
VNM	Normalisation	minmax	65.0	45.0	+20.0	+3.89	-2.22
VNM	Normalisation	rank	55.0	45.0	+10.0	-2.01	-2.22
FPT	Weighting	default	60.0	50.0	+10.0	-1.99	-7.85
FPT	Weighting	equal_weight	60.0	50.0	+10.0	-0.36	-7.85
FPT	Weighting	ic_heavy	65.0	50.0	+15.0	-2.21	-7.85
VNM	Weighting	default	65.0	45.0	+20.0	+6.17	-5.19
VNM	Weighting	equal_weight	50.0	45.0	+5.0	-4.57	-5.19
VNM	Weighting	ic_heavy	60.0	45.0	+15.0	+2.70	-5.19

Table 9: Four-way shared-upstream ablation ($N = 20$ per variant and symbol).*Panel A: Directional decomposition*

Symbol	Full	Alpha-only	Sent.-only	Baseline	$\Delta\alpha$	Δ Sent.	Synergy
FPT	55.0%	50.0%	45.0%	45.0%	+5.0 pp	+0.0 pp	+5.0 pp
VNM	60.0%	50.0%	50.0%	45.0%	+5.0 pp	+5.0 pp	+10.0 pp
VCB	45.0%	55.0%	40.0%	40.0%	+15.0 pp	+0.0 pp	-10.0 pp
Mean	53.33%	51.67%	45.00%	43.33%	+8.33 pp	+1.67 pp	+1.67 pp

Panel B: Compounded account return

Symbol	Full	Alpha-only	Sentiment-only	Baseline
FPT	-4.61%	-7.85%	-7.38%	-6.09%
VNM	-4.03%	+2.26%	-5.86%	-3.96%
VCB	-7.93%	-2.75%	-9.23%	-11.34%

factor pathway provides the more consistent independent directional increment, whereas sentiment is useful only for VNM in this sample. VCB also demonstrates negative interaction: combining sentiment with factors reduces accuracy relative to Alpha-only by 10.0 pp. Economic ordering differs again from directional ordering; for example, VNM Alpha-only is profitable (+2.26%) while its Full variant loses 4.03%. These findings motivate larger-sample interaction tests rather than a universal claim that additional modalities always improve decisions.

7. Limitations & Future Work

7.1. Limitations

L1. Statistical scale and economic interpretation.. The paired benchmark controls upstream reports and evaluates 180 decisions, but each symbol contributes only 20 observations. Moving-block confidence intervals and exact McNemar tests correctly expose this low per-symbol power: no individual McNemar result is significant at 5%, even though the cross-symbol Wilcoxon test is significant. The nine tickers are also not a probability sample of the Vietnamese market. Results should therefore be read as evidence for this benchmark universe rather than a market-wide performance estimate.

Directional skill and economic value remain distinct. The Full System raises mean accuracy by 13.9 pp and improves account return relative to No-Alpha for every symbol, yet its mean compounded return is -1.69% and only three symbols finish positive. Correct SHORT forecasts earn no return because the cash-equity simulator maps SHORT to CASH. Consequently, the reported lift does not establish deployable profitability.

L2. Simplified execution model.. The fixed-cycle ledger removes the previous horizon mismatch: labels, alpha targets, and account returns all use the same next-open-to-target-close net return, wealth is compounded, and fee plus slippage are charged on both BUY and SELL. Nevertheless, the constants are benchmark assumptions. The model omits time-varying bid-ask spreads, taxes, order-book depth, partial fills, volume-dependent market impact, position limits, and portfolio competition among simultaneous symbols. It also assumes that each $S = L = 3$ cycle is closed before the next entry. A venue-calibrated event-driven simulator is required before live deployment.

L3. Alpha selection and sensitivity.. Selecting five factors from an 85-candidate registry creates multiple-testing and selection risks. The point-in-time cutoff prevents future leakage, and the composite score uses $0.40|IC| + 0.35\text{Accuracy} + 0.25\text{Sharpe}$, but short calibration histories can still make signal orientation and ranking unstable. The robustness results confirm this concern: FPT produces a negative lift at $W = 60$, and performance changes materially across normalisation and weighting schemes. Furthermore, translating cross-sectional WorldQuant operators into single-asset time-series operators changes their economic and statistical meaning.

L4. Hosted-model and visual reliability.. Temperature zero and fixed prompts do not guarantee bitwise deterministic output from hosted models. The paired design removes upstream variation within a Full/No-Alpha comparison, but separate runs can still differ because of provider-side model updates or inference nondeterminism. Pattern and Trend agents now compare visual conclusions with deterministic OHLCV rules and downgrade contradictory outputs; this gate reduces obvious chart hallucinations but is not a substitute for an externally labelled pattern-recognition benchmark.

L5. Sentiment coverage and attribution.. Strict cutoff filtering prevents future and undated articles from entering a historical decision. Coverage can nevertheless be sparse and uneven across firms, particularly for smaller or less news-intensive equities. The three-symbol ablation finds a positive independent sentiment contribution only for VNM and a negative factor-sentiment interaction for VCB. This heterogeneity limits any broad causal claim about Vietnamese news sentiment.

7.2. Future Work

F1. Larger and longer evaluation.. The most important extension is a preregistered study over a broader universe such as the VN100 and multiple market regimes. More test points would increase per-symbol power, support sector-level analysis, and permit out-of-time validation rather than repeated analysis of the same recent period.

F2. Stronger baselines and multiplicity control.. Future experiments should add random-walk, logistic, tree-based, and sequence-model baselines on the identical information set. White’s Reality Check, false-discovery-rate control, or the Deflated Sharpe Ratio should accompany factor selection to account for searching across 85 candidates.

F3. Execution and portfolio realism.. A portfolio simulator should model lot sizes, taxes, dynamic spreads, liquidity limits, partial fills, and market impact, and should allocate capital jointly across symbols. Confidence gates and risk-based sizing can then be assessed without conflating forecasting accuracy with all-in binary allocation.

F4. Component reliability.. Repeated hosted-model runs should quantify decision variance. Vision outputs should be compared with deterministic pattern detectors and human-labelled charts, while sentiment should be evaluated against a Vietnamese financial-domain test set. The observed negative VCB interaction also motivates an explicit conflict-resolution mechanism between factor and sentiment evidence.

7.3. Broader Impact

Localized financial AI can broaden research beyond English-language, highly liquid markets, but automated use in a shallow market may amplify noise and herding. The framework should therefore be treated as decision-support research rather than trading advice. Reproducible releases must preserve model and data provenance while respecting news copyright, provider terms, and privacy obligations.

8. Conclusion

This study implemented and evaluated a LangGraph multi-agent system for price-driven directional forecasting in Vietnamese equities. Its principal methodological contributions are a point-in-time 85-factor alpha registry, cutoff-safe Vietnamese sentiment context, shared upstream reports for paired comparison, deterministic consistency checks for vision-agent outputs, and a long-only compounded account ledger whose prediction and execution horizons are aligned.

Across nine symbols and 180 paired test points, the Full System achieved mean accuracy of 65.0%, compared with 51.1% for No-Alpha, yielding a mean lift of +13.9 pp. Eight symbols improved and one tied; the cross-symbol Wilcoxon test gave $p = 0.0078$. The economic evidence is weaker: Full account return exceeded No-Alpha for every symbol, but the mean Full return remained negative at -1.69% , and only HVN, MBB, and MWG were profitable under the stated cost model. These results support improved relative directional performance, not a claim of general trading profitability.

The robustness study confirms that the No-Alpha control is invariant when only alpha normalisation or weighting changes. It also reveals genuine sensitivity: the Full branch ranges from negative to positive lift across window choices. The four-way ablation further shows that OHLCV alpha factors provide the more consistent independent contribution (+8.33 pp on average across FPT, VNM, and VCB), whereas sentiment contributes +1.67 pp on average and interacts negatively with factors for VCB. Thus multimodal evidence should be gated and validated rather than assumed to be additive.

Overall, the experiments demonstrate the value of leakage-safe, paired evaluation for LLM-based financial agents while exposing the gap between directional accuracy and investable returns. Larger universes, longer out-of-time samples, classical baselines, multiplicity-aware factor selection, repeated hosted-model runs, and venue-calibrated execution remain necessary before practical deployment.

CRedit authorship contribution statement

Van-Tang Nguyen: Conceptualization (C); Software (So); Validation (Va); Investigation (I); Original draft preparation (O); Project administration (P); Funding acquisition (Fu).

Hoang-Dung Phan: Methodology (M); Software (So); Formal analysis (Fo); Investigation (I); Resources (R); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Visualization (Vi); Supervision (Su).

Thanh-Tam Do Thi: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Supervision (Su).

Phuong-Thai Nguyen: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Visualization (Vi); Supervision (Su).

Hong-Viet Tran: Conceptualization (C); Methodology (M); Software (So); Validation (Va); Formal analysis (Fo); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Project administration (P).

CONFLICT OF INTEREST STATEMENT (mandatory)

Authors state no conflict of interest.

DATA AVAILABILITY (mandatory)

- The supporting data of this study are openly available from the CafeF at <https://cafef.vn> financial information platform and consist of publicly accessible financial reports of Vietnamese enterprises.

CODE AVAILABILITY

- A demonstration chatbot system developed from the proposed framework is available in the project repository.

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Appendix A. Agent System Prompts

This appendix presents the system and user prompts deployed in the agents.

Note: The original prompts in the codebase are written entirely in Vietnamese to match the target market’s language conventions and to elicit native-level analysis from the LLMs. For the purpose of this paper, they have been translated into English and condensed, while strictly preserving their original logic, constraints, and instructions.

Appendix A.1. Decision Agent Prompt

The Decision Agent receives all upstream reports condensed to save tokens. The prompt enforces a forced binary direction forecast (LONG or SHORT), explicitly distinct from an executed action, and a structured reasoning process. The forecast horizon is dynamically injected as a bar count (e.g., `L=3 bars` for daily data), never as settlement notation. Below is the prompt template (with placeholder variables denoted by `{...}`):

```
You are a Senior Quant Trader with 20 years of experience in the VN stock market.
Asset: **{stock_name}** | Timeframe: **{time_frame}**.
```

```
MARKET RULE: {h_note}
```

```
TIME-INDEX CONTRACT: the input window is [e-W,e); the final visible decision
bar is d=e-1; candidate entry is at the next-bar open O[e]; the immutable
entry is O[e] and the target exit is C[e-1+L]. The label is UP only
when this LONG cycle remains profitable after costs on both legs.
```

```
CORE OBJECTIVE: Deeply evaluate reports from independent AI Agents
to forecast **{h_desc}**. You MUST make a definitive decision:
choose either **LONG** or **SHORT** (NEUTRAL is STRICTLY FORBIDDEN).
```

```
---
```

```
### [1] TREND ANALYSIS (Trend Agent)
{trend_report}
```

```
---
```

```
### [2] PRICE PATTERNS (Pattern Agent)
{pattern_report}
```

```
---
```

```
### [3] TECHNICAL INDICATORS (Indicator Agent)
{indicator_report}
```

```
---
```

```
### [4] ALPHA FACTORS & ORDER FLOW (Alpha Agent)
{alpha_report}
```

```
### [5] NEWS & MARKET SENTIMENT (Sentiment Analysis)
{sentiment_report}
```

```
---
```

```
## REASONING GUIDE (Chain of Thought)
1. Market Context (Trend)
2. Price Action & Timing (Pattern)
3. Momentum (Indicator)
4. Flow & Sentiment (Alpha + Sentiment)
```

```
## MANDATORY OUTPUT FORMAT
```json
{
 "decision": "<LONG or SHORT>",
 "forecast_horizon": "{h_val}",
```

```

"confidence": "<Very High | High | Medium | Low>",
"risk_reward_ratio": <float>,
"evidence_for": "<3 sentences summarizing reasons>",
"evidence_against": "<Biggest risk>",
"justification": "<Brief justification>"
}
'''

```

The agent reports 4 (when Alpha Agent is active) or 3 (No-Alpha ablation) upstream summaries. Sections [4] and [5] are omitted entirely when `include_alpha=False`.

### *Appendix A.2. Pattern Agent Prompt*

The frozen configuration names the vision-language model `qwen/qwen3.8-27b`; the following Pattern Agent prompt is prototype source, not a provenance-validated research artifact:

```

/no_think
You are an AI specializing in Price Action pattern analysis.

ABSOLUTE RULES:
1. ONLY return the 6 fields below. Do NOT add any other fields.
2. NO reasoning, NO explanations.
3. NO fallback scenarios. Provide exactly ONE definitive assessment.
4. Each field on ONE line. Be concise.

MANDATORY FORMAT:
Pattern: <Name>
Confidence: High | Medium | Low
Forecast Bias: Bullish | Bearish | Neutral
Evidence: <Brief description>
Key Candle: <List key candle>
Trade Implication: <Scenario>

```

The 16 reference patterns provided to the model include: Inverse Head and Shoulders, Double Bottom, Falling/Rising Wedge, Triangle, Bull/Bear Flag, Rectangle, V-Reversal, and Clear Trend.

If the vision model fails, the agent falls back to a text-only analysis of the most recent 20 OHLCV rows using the text LLM (`openai/gpt-oss-20b`).

### *Appendix A.3. Trend Agent Prompt*

The Trend Agent also utilizes a vision-language model to evaluate dynamically generated support and resistance lines.

You are an expert at analyzing K-line trend patterns in high-frequency trading.  
Your sole objective: PREDICT {h\_desc}.  
You must always reply in the exact markdown format requested.  
NEVER write all fields on a single line.

This is a {time\_frame} candlestick chart with automatic trendlines:

- **Blue line** = support line (derived from close prices)
- **Red line** = resistance line (derived from close prices)

Your tasks:

1. Analyze how price interacts with the blue and red lines.
2. Determine if candles are bouncing, breaking out, or compressing.
3. Evaluate the slope and distance between the two lines.
4. Predict the direction for **{h\_val}**: bullish, bearish, or sideways.

MANDATORY FORMAT:

**Trend Direction:** Bullish | Bearish | Sideways  
**Support Level:** <specific price>  
**Resistance Level:** <specific price>  
**Trendline Slope:** Rising | Falling | Flat  
**Price vs Support:** <bouncing / breaking / compressing>  
**Detailed Analysis:** <2-3 sentences of price action analysis>  
**Trend Prediction:** <1-2 sentences forecasting {h\_desc}>  
**Confidence:** High | Medium | Low

#### Appendix A.4. Indicator Agent Prompt

The Indicator Agent receives a pre-calculated table of technical indicators and their deterministic classifications (computed in Python). Its prompt forces it to adhere strictly to the system's hardcoded signals rather than re-interpreting the raw data.

You are an expert at High-Frequency Trading technical analysis.  
SOLE OBJECTIVE: Predict price direction for **{h\_desc}**.

MOST IMPORTANT RULES:

1. SIGNAL CLASSIFICATION and CONSENSUS have already been determined by the system - you MUST use these exact conclusions. Do NOT count or change them.
2. Each indicator has a strict [END ... SECTION] boundary. Do NOT use numbers from one section to describe another.
3. Your only task: write a clear, actionable qualitative interpretation for {h\_short}.
4. NO hallucinated numbers. NO saying 'missing information'.

## Data Table (Calculated & Classified)

[Pre-rendered Table and System Consensus injected here]

```

Requirement: Write an interpretation report
For EACH indicator, use this exact template:
[Indicator Name]
- {h_short} Signal: [EXACT system classification]
- Key levels/crossovers: [Specific numbers from the correct section]
- Momentum: [accelerating / decelerating / stable] - short explanation

Then summarize the final consensus:
Signal Convergence Summary - Prediction for {h_short}
- Bullish agreement: **{n_tang}/{total}**
- Bearish agreement: **{n_giam}/{total}**
- Primary Trend: **{consensus}** - Confidence: **{confidence}**
- Conclusion: [1-2 sentences summarizing potential movement]

```

#### *Appendix A.5. Sentiment LLM Prompt (Subroutine of Alpha Agent)*

Sentiment Analysis is invoked programmatically as a subroutine within the Alpha Agent node rather than acting as a standalone agent in the DAG. The configured hosted ViSoBERT endpoint is the primary scorer and a deterministic lexicon is the explicit fallback. The actual scorer is retained per article; the text LLM only synthesizes a qualitative summary of the resulting news bias and related-company impacts.

```

You are an expert at analyzing sentiment in the VN stock market.
Synthesize data from press articles. Reply in markdown.

Analyze sentiment for **{target}** ({time_frame}).

=== AGGREGATE RESULTS ===
Article Count: {article_count} | Avg Score: {avg_score} | Label: {label}
Distribution: {positive} / {neutral_count} / {negative}
Sentiment Bar: {sentiment_bar_visual} | Model: {model_note}

=== LATEST 15 ARTICLES ===
{art_sum}

=== RELATED COMPANIES ===
{rel_sum}

Write the report using this template:

Sentiment Summary - {target}
Main Takeaway: [1-2 sentences]
Detailed Analysis:
- News Trend: [...]

```

- Positive Signals: [...]
- Negative Signals: [...]
- Pressure from Related Companies: [...]

### Appendix A.6. Alpha Agent LLM Reasoning Prompt

After computing the five alpha factors programmatically, the Alpha Agent invokes the text LLM to synthesise a qualitative interpretation. The prompt template is:

```
You are a quantitative and order flow analyst forecasting {horizon_label}.
Below are OHLCV-only alpha factors and a separate sentiment context
for {stock_name}.

ALPHA FACTORS RESULTS
{report_md}

SEPARATE SENTIMENT & NEWS CONTEXT
{sentiment_context_md}

FUSION CONTRACT:
- Alpha values and ranks use OHLCV only.
- Sentiment does not enter factor formulas; it is text context only.

Compare the two evidence sources qualitatively to:
1. Provide a short comment (1-2 sentences) on current momentum.
2. Synthesize: Are flow and sentiment in agreement or contradiction?
3. Identify hidden risks/opportunities from news not captured by technical alphas.

Do NOT over-analyze. Deduce from the numbers to conclude the view for {horizon_label}.
```

## Appendix B. LangGraph State Schema

The entire multi-agent pipeline shares a single typed state object defined as a Python `TypedDict`. Table B.10 lists all fields, grouped by the agent that writes them and annotated with their data types. The backtest wrapper records the zero-based provenance separately for each test point: exclusive boundary  $e$ , window  $[e - W, e)$ , decision  $d = e - 1$ , entry  $e$ , target  $e - 1 + L$ , and the source timestamp for every index. These fields make calendar gaps (including weekends and exchange holidays) explicit rather than inferring timestamps by adding calendar days.

*Graph topology.* The DAG is constructed in `graph_setup.py` using the LangGraph `StateGraph` API. In the Full System configuration, the sequential edge order is:

START → Indicator → Alpha → Pattern → Trend → Decision → END

Table B.10: Complete LangGraph state schema (**IndicatorAgentState**)

Field	Type	Description
<i>Input fields (set before graph execution)</i>		
kline_data	dict	OHLCV dictionary for technical indicator computation
interval	str	Canonical market-data interval code (e.g., "1d")
forecast_horizon_bars	int	Target horizon $L$ ; label at decision $d = e - 1$ is $\log(C_{e-1+L}/C_{e-1})$
holding_policy	str	Explicit next-bar-open-to-target-close execution policy
alpha_history	dict	Point-in-time OHLCV prefix ending at decision cutoff; no entry/target bars
alpha_cutoff	str	Exact timestamp of the final observable decision bar
history_requirements	dict	Separate chart and alpha-calibration requests; calendar lookback and actual-bar minimums are distinct
window_end_date	str	Decision cutoff timestamp used for point-in-time sentiment availability
time_frame	str	Time period for K-line data (e.g., "1 ngày")
stock_name	str	Stock ticker symbol
is_backtest	bool	Flag to block external crawlers in backtest mode
pattern_image	str	Base64-encoded candlestick chart PNG
trend_image	str	Base64-encoded trendline-annotated chart PNG
<i>Indicator Agent outputs</i>		
rsi	List[float]	RSI values (period 14)
macd, macd_signal, macd_hist	List[float]	MACD line, signal, histogram (12, 26, 9)
stoch_k, stoch_d	List[float]	Stochastic %K, %D (14, 3, 3)
roc	List[float]	Rate of Change (period 10)
willr	List[float]	Williams %R (period 14)
bb_upper, bb_middle, bb_lower	List[float]	Bollinger Bands (20, 2)
adx, plus_di, minus_di	List[float]	ADX and directional indicators
indicator_report	str	Markdown summary with consensus
<i>Alpha Agent outputs</i>		
alpha_report	str	Five alpha factors + LLM reasoning
<i>Pattern Agent outputs</i>		
pattern_report	str	Six-field markdown pattern analysis
<i>Trend Agent outputs</i>		
trend_report	str	Eight-field markdown trend analysis
<i>Sentiment outputs (written by Alpha Agent internally)</i>		
sentiment_report	str	CafeF sentiment summary
sentiment_data	Dict[str, Any]	Display articles plus complete compact cutoff-safe observations
sentiment_norm	Dict[str, Any]	Time-decayed sentiment, relative context, availability and provenance
<i>Decision Agent outputs</i>		
final_trade_decision	str	Raw LLM response containing JSON decision
decision_prompt	str	Full prompt sent to the Decision Agent
messages	List[BaseMessage]	LangChain message history

During evaluation, graph execution is split into a variant-independent upstream graph and variant-specific decision graphs. Indicator, Pattern, and Trend run once per exact cutoff; the resulting state is deep-copied into Full, Alpha-only, Sentiment-only, and Baseline branches as required by the experiment. This shared-artifact topology prevents Alpha settings from changing the No-Alpha evidence snapshot.

## Appendix C. Proprietary Alpha Formula Implementation

This appendix provides the exact Python implementation of the five proprietary alpha formulas listed in Table 2. All formulas operate on the technical variables extracted from the OHLCV window by the function `_extract_tech_vars()`.

### Appendix C.1. $\alpha_1$ : Flow-Driven Momentum (FDM)

```
def _alpha1_fdm(tv: dict) -> dict:
 roc1_adj = tv["roc1_adjusted"] # ROC(1) / rolling_std(ROC, 20)
 vol_surge = tv["vol_surge"] # Volume / ADV(20)
 raw = roc1_adj * vol_surge * 0.5
 value = math.tanh(raw)
```

# Signal threshold: |value| > 0.10

Where `roc1_adjusted` is the one-period rate of change normalised by its 20-period rolling standard deviation, and `vol_surge` is the ratio of current volume (or intra-bar range proxy when volume is unavailable) to the 20-period average volume.

#### *Appendix C.2. $\alpha_2$ : MACD-Flow Proxy (MFP)*

```
def _alpha2_mfp(tv: dict) -> dict:
 roc1_adj = tv["roc1_adjusted"]
 macd_hist = tv["macd_hist_cur"]
 z_macd = tanh(macd_hist / (close * 0.005)) * 1.5
 raw = z_macd + roc1_adj * 0.5
 value = math.tanh(raw)
```

This factor always uses price-derived MACD histogram and ROC inputs. Sentiment coverage does not switch or modify its formula; sentiment is provided only to the later LLM text-context comparison.

#### *Appendix C.3. $\alpha_3$ : Liquidity Void Reversion (LVR)*

```
def _alpha3_lvr(tv: dict) -> dict:
 z_close_dev5 = tv["zscore_close_dev5"]
 raw = -1.0 * z_close_dev5 * 0.8
 value = math.tanh(raw)
 # Signal threshold: |value| > 0.15 (higher than other alphas)
```

Where `zscore_close_dev5` is the z-score of the series  $\frac{C_u - \text{SMA}_5}{C_u}$  over a 20-period rolling window at bar position  $u$ . The negative sign creates a mean-reversion signal: when the close is abnormally far above SMA(5), the alpha predicts a pullback.

#### *Appendix C.4. $\alpha_4$ : Bollinger Squeeze & Flow (BFE)*

```
def _alpha4_bfe(tv: dict) -> dict:
 bb_pct_b = tv["bb_pct_b"] # Bollinger %B
 vol_z5 = tv["vol_zscore_5"] # Volume z-score (5-period)
 raw = (bb_pct_b - 0.5) * vol_z5 * 1.5
 value = math.tanh(raw)
```

This formula fires when price breaks above/below the Bollinger midline ( $\%B \neq 0.5$ ) with confirming volume expansion ( $|Z_{\text{vol}}| > 0$ ). The multiplicative interaction ensures both conditions must co-occur.

#### *Appendix C.5. $\alpha_5$ : Order Flow Exhaustion (OFE)*

```
def _alpha5_ofe(tv: dict) -> dict:
 roc1_adj = tv["roc1_adjusted"]
 close_pos = tv["close_pos"] # (Close - Low)/(High - Low) - 0.5
 raw = roc1_adj * close_pos * 3.0
 value = math.tanh(raw)
```

Where `close_pos`  $\in [-0.5, 0.5]$  measures where the closing price falls within the intra-bar range, centred at zero. A positive `close_pos` with positive momentum confirms buying pressure; the  $\times 3.0$  amplification increases signal responsiveness.

## Appendix D. Indicator Signal Classification Logic

Section 4 describes the Indicator Agent’s deterministic signal classification. This appendix provides the exact classification logic implemented in Python (function `_classify_signals`), which is injected into the LLM prompt as immutable ground truth.

### Appendix D.1. MACD Classification

1. **Priority 1 — Crossover detection:** If  $\text{MACD}_{t-1} < \text{Signal}_{t-1}$  and  $\text{MACD}_t \geq \text{Signal}_t$ , classify as BULLISH (Golden Cross). Conversely, classify as BEARISH (Death Cross).
2. **Priority 2 — Histogram direction:** If no crossover occurred, classify based on histogram sign and trend (expanding or contracting).

### Appendix D.2. RSI Classification

- $\text{RSI} \leq 30$ : BULLISH (oversold bounce expected)
- $\text{RSI} \geq 70$ : BEARISH (overbought pullback expected)
- $55 \leq \text{RSI} < 70$ : BULLISH (positive momentum)
- $30 < \text{RSI} \leq 45$ : BEARISH (negative momentum)
- Otherwise: NEUTRAL

### Appendix D.3. Stochastic Oscillator Classification

1. **Priority 1 — %K/%D crossover:** %K crossing above %D  $\rightarrow$  BULLISH; %K crossing below %D  $\rightarrow$  BEARISH.
2. **Priority 2 — Extreme zones:**  $\%K \geq 80 \rightarrow$  BEARISH (overbought);  $\%K \leq 20 \rightarrow$  BULLISH (oversold).
3. **Otherwise:** Compare current %K vs %D position.

### Appendix D.4. Williams %R Classification

- $\%R \geq -20$ : BEARISH (overbought, near 0)
- $\%R \leq -80$ : BULLISH (oversold, near  $-100$ )
- Neutral zone: Classify based on 3-bar trend direction (improving toward 0  $\rightarrow$  BULLISH; deteriorating toward  $-100 \rightarrow$  BEARISH).

### Appendix D.5. ROC Classification

- $\text{ROC} > +0.5\%$ : BULLISH
- $\text{ROC} < -0.5\%$ : BEARISH
- Otherwise: NEUTRAL

### Appendix D.6. Consensus Rule

The overall consensus is determined by majority vote over the five indicators:

- $\geq 4$  agreement  $\rightarrow$  High confidence
- 3 agreement  $\rightarrow$  Medium confidence
- Otherwise  $\rightarrow$  Low confidence

## Appendix E. Software Dependencies and Reproducibility

Table E.11 lists the core software dependencies used in the implementation. All packages are specified in `requirements.txt` and are installable via `pip`.

Table E.11: Software dependencies

Package	Version	Purpose
flask	latest	Web interface for real-time demo
pandas	latest	Data manipulation and OHLCV processing
numpy	latest	Numerical computation
matplotlib	latest	Chart rendering backend
mplfinance	latest	Candlestick chart generation
scipy	latest	Statistical functions
TA-Lib	latest	Technical indicator computation (MACD, RSI, etc.)
langchain	latest	LLM abstraction layer
langchain-groq	latest	Groq API integration for LLM inference
langgraph	latest	Multi-agent DAG orchestration framework
vnstock	$\geq 3.0.0$	Vietnamese stock market data (VCI/MSN sources)
requests	latest	HTTP client for CafeF crawling and hosted sentiment inference
beautifulsoup4	latest	HTML parsing for article content extraction
Pillow	latest	Image processing utilities

*Sentiment scorer provenance..* The runtime does not load `BertTokenizer`, `transformers`, or `torch`. It sends at most 512 text characters to the hosted Hugging Face Inference API route configured for `5CD-AI/Vietnamese-Sentiment-visobert`; an unavailable or failed route falls back to the deterministic lexicon implemented in the same module. The supplied cache schema stores the resulting label, numeric score, and confidence, but not the actual scorer, model identifier, or fallback flag. The audit therefore does not infer provenance from those numerical fields. Every such record is classified as `legacy-unknown`; a record contributes to the verified ViSoBERT or verified lexicon count only when it contains explicit scorer metadata. This conservative rule prevents an endpoint configuration from being reported as evidence that the endpoint successfully scored a particular article.

*Reproducibility artifacts..* The following artifacts are provided or specified for reproducibility:

- **Model versions:** The reported text default is `openai/gpt-oss-20b`; the vision default is `qwen/qwen3.8-27b`. Both use the Groq API. The backtest engine resolves these configurations directly and stores the effective run settings with its result artifacts.
- **Temperature:** The current default is  $T = 0.0$ . Effective temperature, token caps, reasoning format, retry count, and a stable credential-free configuration hash are recorded per run; this metadata does not guarantee deterministic hosted-model output.
- **Retry policy:** The application policy defaults to at most three text calls (two retries with 5/10-second exponential waits) and two vision calls (one retry with a 3-second wait) when SDK retries are disabled. Positive SDK `max_retries` budgets make the SDK the single retry owner

Table E.12: Ticker-selection strata and point-in-time sentiment coverage.

<b>Ticker</b>	<b>Sector</b>	<b>Design stratum</b>	<b>Dated/total</b>	<b>V/L/U</b>	<b>Mean 90d</b>	<b><math>\geq 8</math> points</b>
BHN	Food/beverages	State-associated, specialised	20/20	0/0/20	1.15	0/20
CMG	Information technology	Mid-cap	20/20	0/0/20	0.85	0/20
FPT	Information technology	VN30/large-cap	20/20	0/0/20	1.45	2/20
HVN	Aviation and tourism	State-associated	20/20	0/0/20	9.45	19/20
MBB	Banking	VN30/large-cap	20/20	0/0/20	7.35	7/20
MWG	Retail	VN30/large-cap	20/20	0/0/20	6.05	8/20
VCB	Banking	VN30/large-cap	20/20	0/0/20	9.90	18/20
VJC	Aviation and tourism	Large private carrier	20/20	0/0/20	10.10	17/20
VNM	Consumer food	VN30/large-cap	20/20	0/0/20	5.60	2/20
<b>All</b>	<b>Nine-symbol purposive sample</b>		<b>180/180</b>	<b>0/0/180</b>	<b>5.77</b>	<b>73/180</b>

*Notes:* V/L/U denotes verified ViSoBERT, verified lexicon, and unknown scorer counts. “Mean 90d” is the mean number of dated articles in the inclusive 90-day lookback at the 20 test cutoffs. The final column counts test points with at least eight eligible articles. The underlying machine-readable audit is generated by `scripts/analyze_sentiment_coverage.py`.

and reduce the application wrapper to one call; SDK-internal attempts are not observable and remain unknown in telemetry. Inter-test pacing is recorded separately from retry delays.

- **Historical sentiment cache:** Nine JSON files (`sentiment_cache_{SYMBOL}.json`), one per evaluated stock, contain pre-crawled and scored CafeF articles. Point-in-time filtering uses the parsed publication date; undated records are rejected in strict research mode. The supplied files lack scorer metadata, so all 180 cached scores are labelled `legacy-unknown` in the coverage audit.
- **Default configuration:** Stored in `default_config.py` with all hyperparameters.

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